AUCKLAND CRASH ANALYSIS
OVERVIEW 2015-2025 NZTA CRASH DATASET

Severity: All

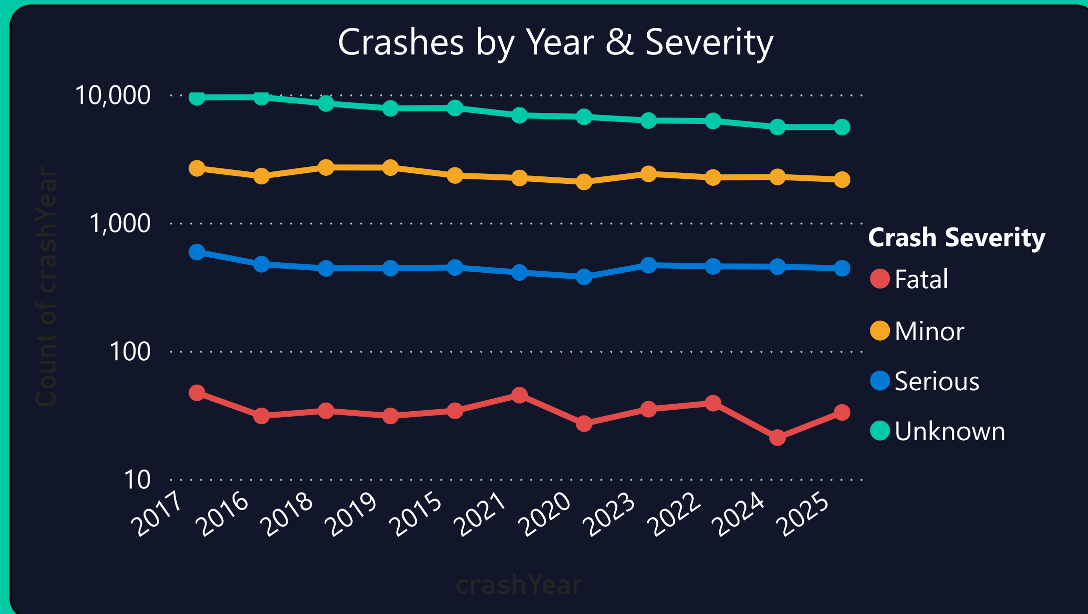
Year: All

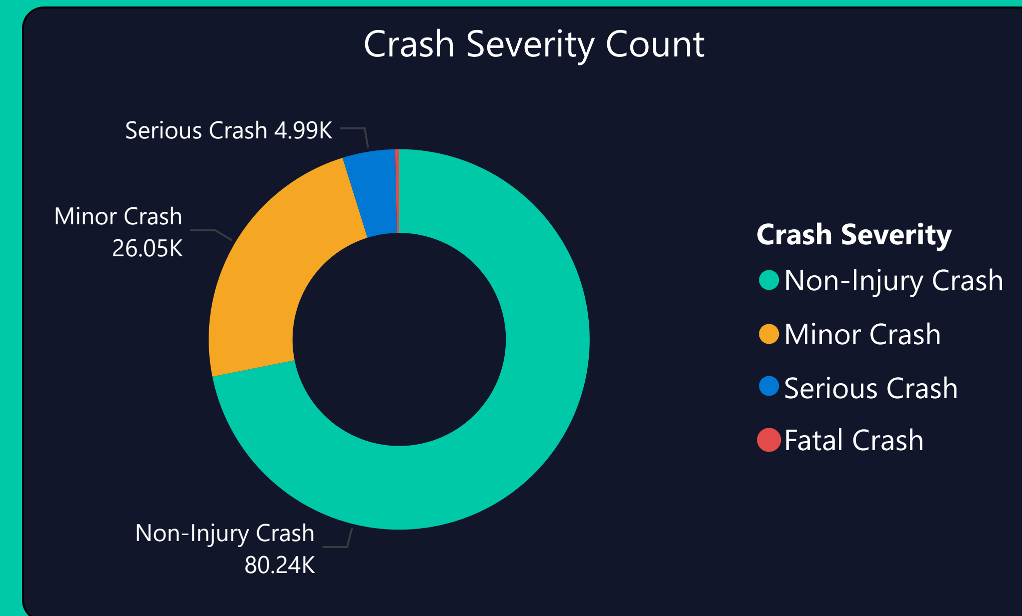
112K Total Crashes

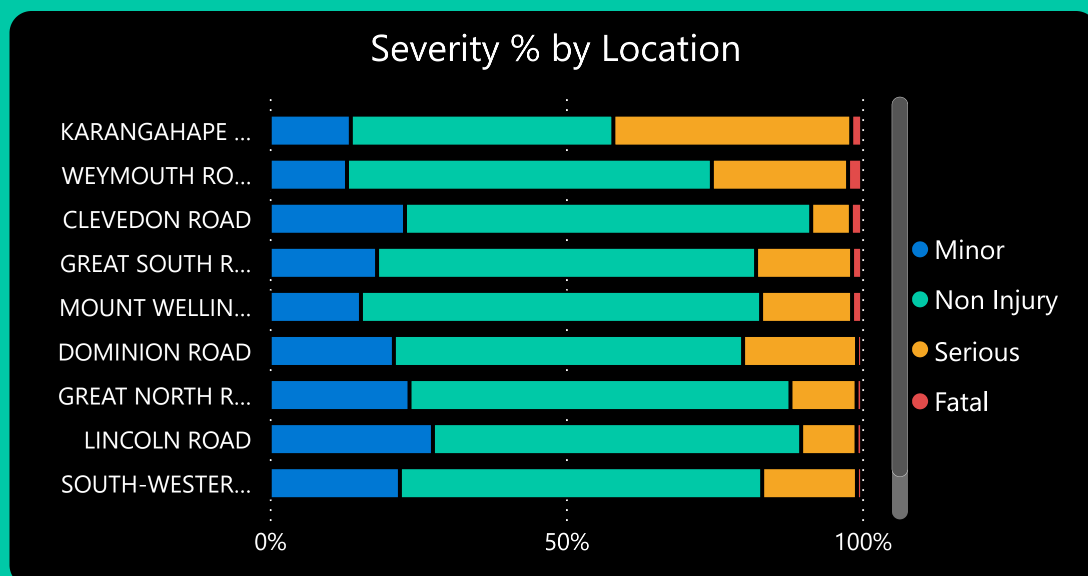
377 Fatal Crashes

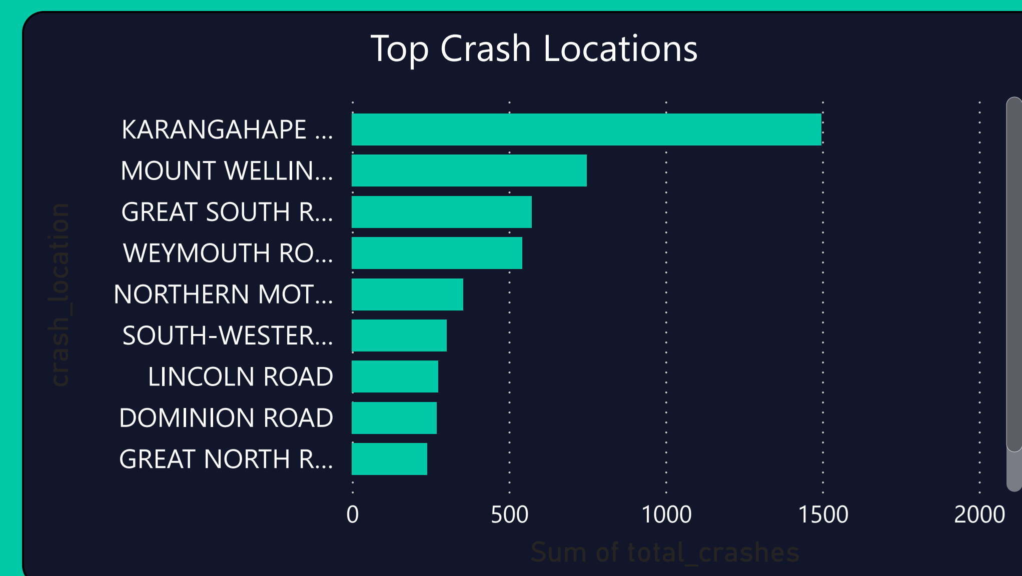
0.73 Best Model...

4.80 Severity Rat...









SPATIAL HOTSPOT ANALYSIS GI STATISTICS - GETIS ORD - AUCKLAND REGION

KEY INSIGHTS -Total crash coverage is the primary Gi* metric. Severe crash coverage is lower because severe crashes (4.8% of all crashes) follow a more dispersed spatial

16.75
Hotspot Coverag...

14.49
Severe Hotspo...

0.62
Morans I

36
Hotspot

HOTSPOT MAP

Hot Spot 95% Hot Spot 99%

YEAR
All

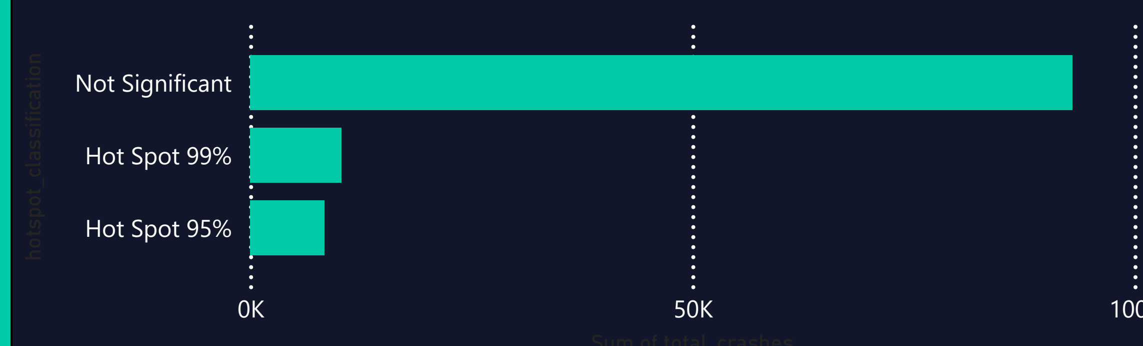
HOTSPOT

Hot Spot 95% | Hot Spot 99% | Not Significant

TOP HOTSPOT GRID CELL

grid_lat	grid_lon	total_crashes	severe_crashes	gi_star_z	gi_star_p	hotspot_classification
-37.03	174.91	328	19	2.02	0.00	Hot Spot 95%
-37.01	174.89	264	16	1.97	0.00	Hot Spot 95%
-37.00	174.88	552	17	2.18	0.00	Hot Spot 95%
-37.00	174.89	763	21	2.46	0.00	Hot Spot 95%
-37.00	174.90	353	17	2.08	0.00	Hot Spot 95%
-36.99	174.87	284	7	2.23	0.00	Hot Spot 95%
-36.99	174.88	587	18	2.70	0.00	Hot Spot 99%
-36.99	174.89	715	34	2.57	0.00	Hot Spot 95%
-36.99	174.90	359	17	2.01	0.00	Hot Spot 95%

CRASHES BY HOTSPOT CLASSIFICATION



PREDICTIVE SEVERITY MODEL
LOGISTIC REGRESSION · RANDOM FORREST · XGBOOST

PRED PROB...

All

YEAR

All

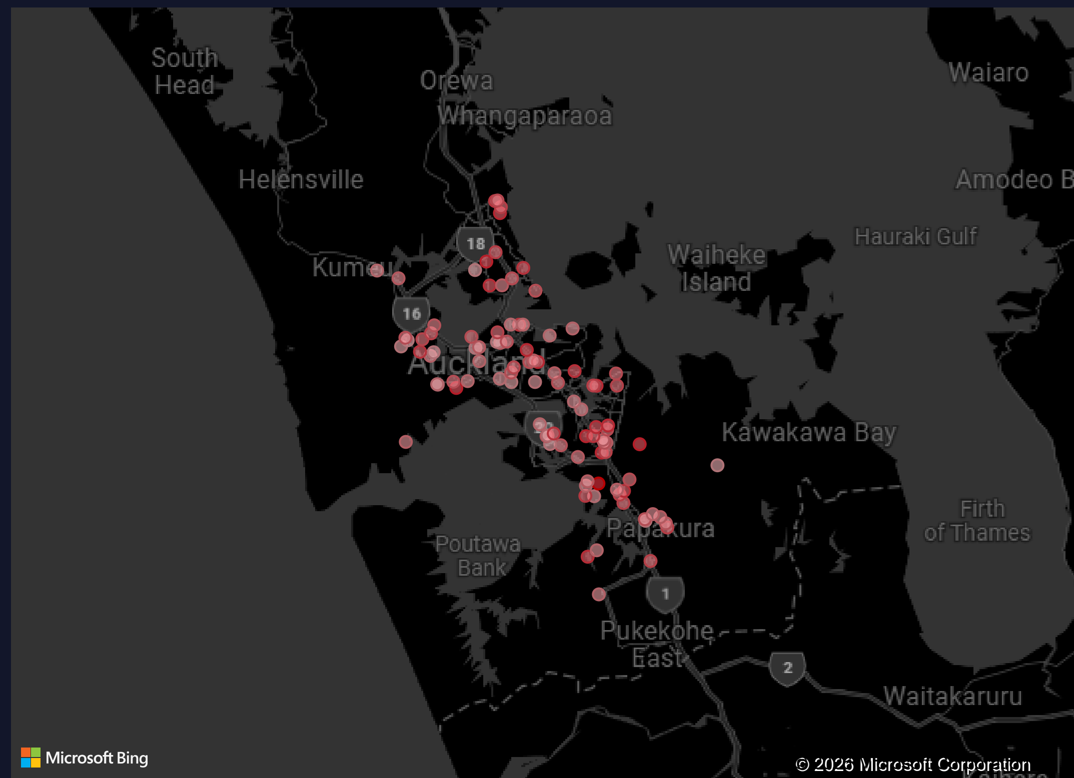
0.73
Best Model AUC

1K
High Risk Crashes

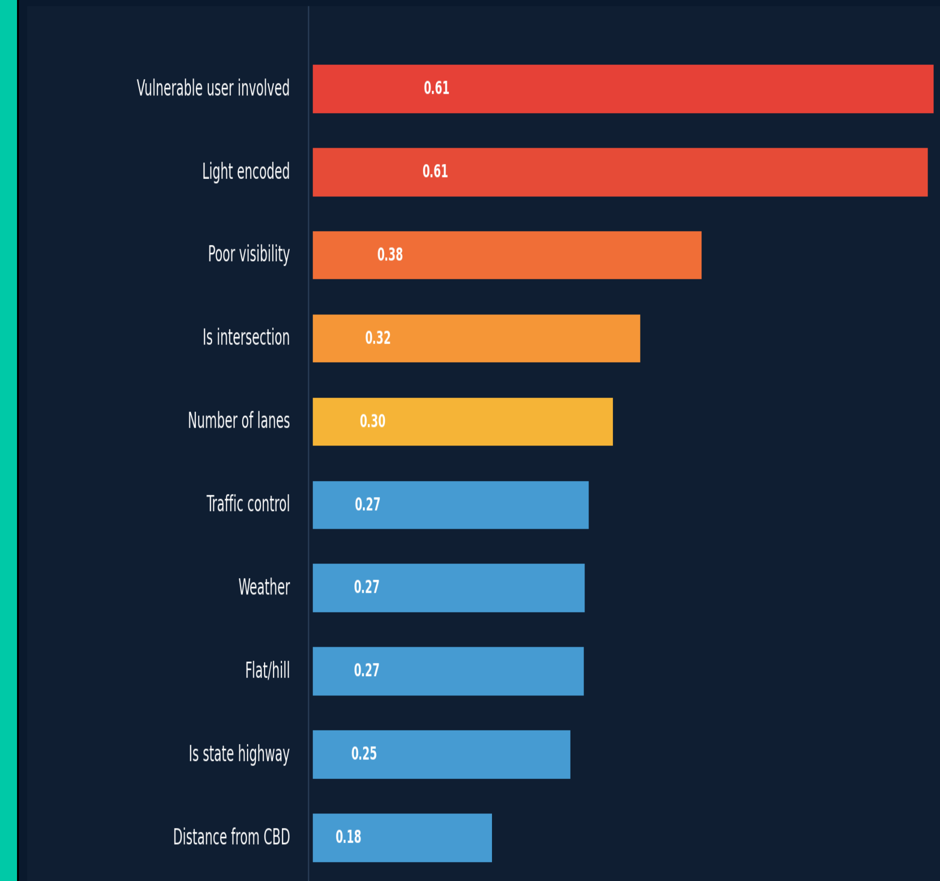
0.64
Best Recall Model

KEY INSIGHTS
ROC-AUC measures the model's ability to rank severe crashes above non-severe ones. Public CAS data does not include crash hour or driver behaviour variables, which limits maximum achievable AUC. The

HIGH-RISK CRASH LOCATIONS (Pred. Prob ≥ 0.90)



FEATURE IMPORTANCE (Mean |SHAP| value)



MODEL PERFORMANCE COMPARISION

Model	ROC-AUC	Recall	Precision	F1	Improvement_pct
Logistic Regression	0.73	0.58	0.12	0.20	0.00
Random Forest	0.71	0.43	0.16	0.23	-2.54
XGBoost	0.72	0.64	0.08	0.15	-1.85
Total					

ROC-AUC BY MODEL

TEMPORAL & HOLIDAY ANALYSIS

ANNUAL TRENDS · HOLIDAY PERIOD SEVERITY · 2015-2025

KEY INSIGHTS

The holiday–severity association is statistically significant (p=0.018) but the effect size is negligible (Cramer's V=0.0095). Labour Weekend shows the highest severity rate (+1.47pp above baseline) suggesting targeted enforcement, not calendar-wide resource reallocation, is the

0.02

Holiday Chi2 p-...

6.27

Labour Weeken...

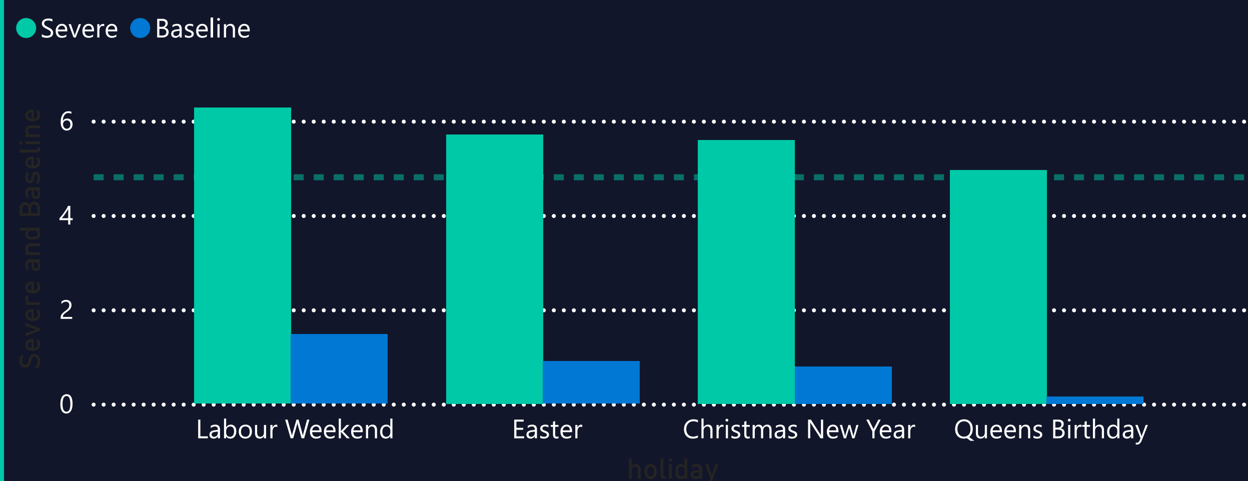
slope_crashes_per_...

-418.17

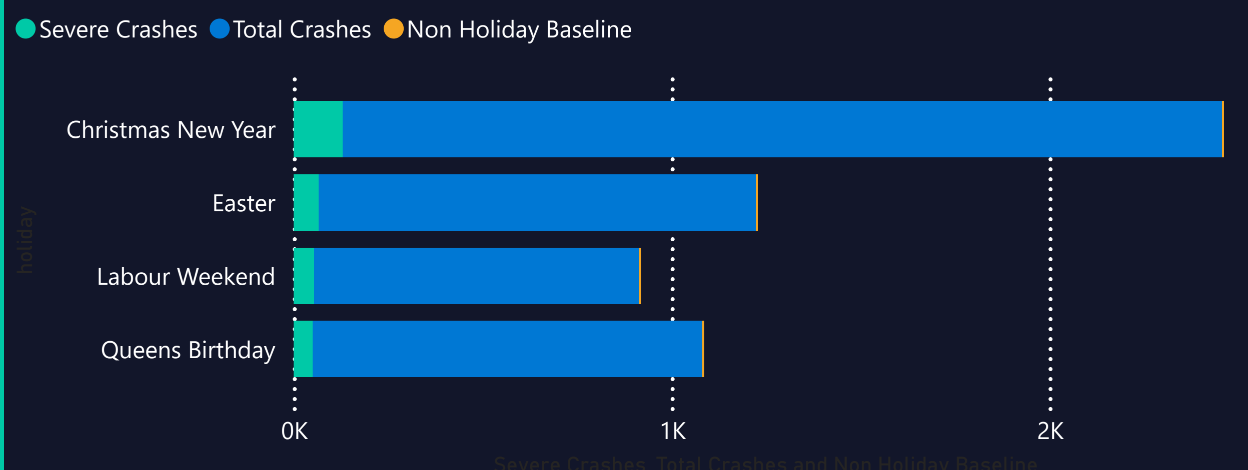
cramers_v_holiday

0.01

HOLIDAY SEVERITY VS BASELINE



HOLIDAY CRASH VOLUME VS BASELINE



AREA

All

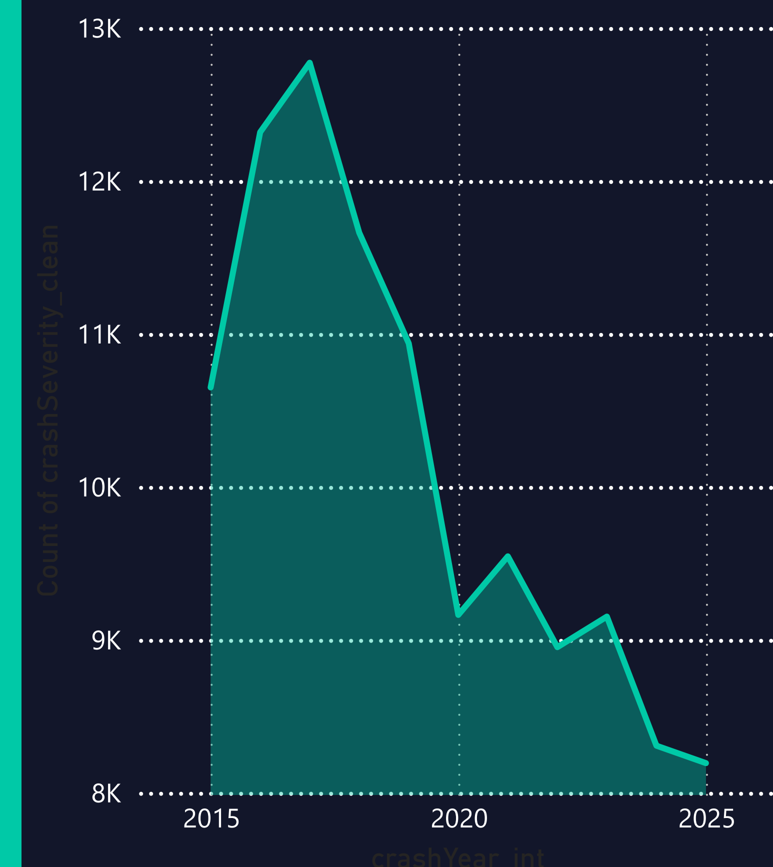
YEAR

All

HOLIDAY

All

ANNUAL CRASH TREND



VULNERABLE USER ANALYSIS

PEDESTRAINS · CYCLIST · MOTORCYCLIST VS VEHICLE ONLY

KEY INSIGHTS

Crashes involving vulnerable road users are 8.95× more likely to result in fatal or serious injury than crashes involving vehicle occupants only (23.7% vs 2.6% severe rate). Road lane type, lighting conditions, and urban classification are the strongest

User Type

- Cyclist
- Motorcyclist

8.95

VU Relative...

23.33

VU Severe...

2.53

Non-VU...

10

Significant...

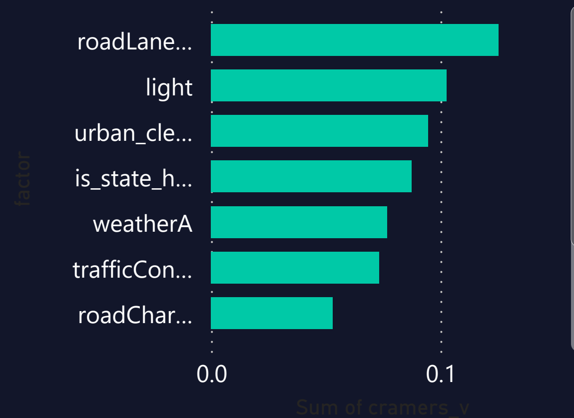
LOCATION AND CRASH COUNT

Location	Severe	Crash Count
GREAT SOUTH ROAD	101	432
GREAT NORTH ROAD	76	266
DOMINION ROAD	42	227
SOUTHERN MOTORWAY	41	184
SH 1N	37	224
MANUKAU ROAD	32	121
TAMAKI DRIVE	31	139
PAKURANGA ROAD	25	70
Total	**2848**	**12210**

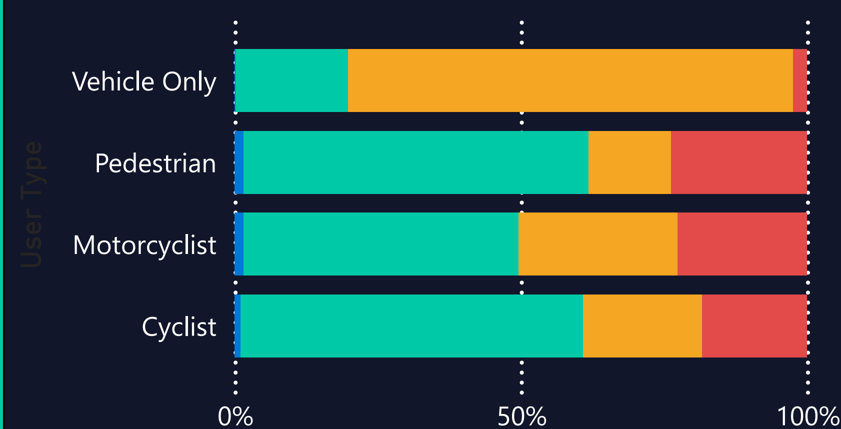
RISK FACTOR SIGNIFICANCE

Factor	P value	Cramers v	Significant
roadLane_clean	0.00	0.13	True
light	0.00	0.10	True
urban_clean	0.00	0.09	True
is_state_highway	0.00	0.09	True
weatherA	0.00	0.08	True
trafficControl	0.00	0.07	True
roadCharacter	0.00	0.05	True
flatHill_clean	0.00	0.04	True
Total	**0.00**	**0.68**	

RISK FACTOR BY CRAMERS V



CRASH SEVERITY BY USER TYPE



CRASH COUNT VS LOCATION VS USER TYPE